00-D-1297B/GEN
April 5, 1974
SUPERSEDING
Fed. Spec. 00-D-1297A
July 27, 1970
(see 6.5)

FEDERAL SPECIFICATION

DRYING AND CONDITIONING TUMBLERS, LAUNDRY, COMMERCIAL: GENERAL SPECIFICATION FOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for Drying and Conditioning Tumblers. Detail requirements for a particular tumbler and size are covered by the applicable detail Specification 00-D-1297/1 through 00-D-1297/5.

1.2 Classification.

1.2.1 Types, styles, reversibility, sizes, and kind of heat. The type, style, reversibility, size, and kind of heat shall be as designated in the detail specification for the tumbler specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CC-M-636	-	Motor, Alternating Current (Fractural Horsepower).
CC-M-641	-	Motors, Alternating Current (Integral Horsepower, 200 hp and Smaller).
QQ-C-390	-	Copper Alloy Castings (Including Cast Bar).
QQ-N-281	-	Nickel-Copper-Alloy-Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288	-	Nickel-Copper-Alloy and Nickel-Copper-Silicon Alloy Castings.
DDD-T-551	-	Towel, Bath, Cotton, Terry.

FSC 3510

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifcations and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-L-3153 - Laundry and Dry-Cleaning Machinery and Equipment (for Fixed Installations), Preparation for Delivery for Domestic and Overseas Shipment.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement Of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

29 CFR 1910 - Occupational Safety and Health Act (OSHA) Standards

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:00Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturers' Association (NEMA) Standards Publication

ICS - Industrial Controls and Systems (1970)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

Underwriters' Laboratories, Inc. (UL) Standards

(Application for copies of applicable standards should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, LI, New York 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Fire Protection Association (NFPA) Standard

No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

American Society for Testing and Materials (ASTM) Specifications

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A176 - Stainless and Heat-Resisting Chromium Steel, Plate, Sheet and Strip

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Factory Insurance Association (FIA)

(Application for copies should be addressed to Factory Insurance Association, 85 Woodland Street, Hartford, CT 06102.)

Factory Mutual Engineering Associates (FMTL)

(Application for copies should be addressed to Factory Mutual Testing Laboratories, 1151 Providence Highway, Norwood, MA 02062.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:00Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be as specified herein and in accordance with applicable detail specification.

3.2 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.4 and 6.3).

3.3 Standard product. The tumblers delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof shall be new and interchangeable.

3.4 Codes and standards. The tumblers shall comply with the applicable requirements of NFPA No. 70, NEMA, FIA, and OSHA.

3.4.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfacroty evidence that the tumbler that he proposes to furnish under this specification meet the applicable requirements of NFPA No. 70, UL, NEMA, ICS-70, FIA, FMTL and OSHA, 29 CFR-1910.

3.4.1.1 NFPA. Acceptable evidence of meeting requirements of NFPA standards shall be the manufacturer's certified statement that the tumbler is wired for use in non-hazardous locations in accordance with NFPA Standard No. 70.

3.4.1.2 UL. Acceptable evidence of meeting requirements of UL shall be the UL label, a listing mark, or a certified test report from a recognized testing laboratory, acceptable to the Government indicating that the tumbler wiring, fittings and conduit conform to applicable UL requirements.

3.4.1.3 NEMA. Acceptable evidence of meeting the requirements of NEMA standards shall be the manufacturer's certified statement that the motor controllers, switches, relays, and time delays, conform to NEMA Standard Publication No. ICS-1970. A tag or label attached to these components indicating compliance to this standard is acceptable evidence.

3.4.1.4 FIA and FMTL. Acceptable evidence of meeting the requirements of FIA or FMTL shall be the manufacturer's certified statement that the tumbler burners valving for gas, oil, propane, butane or gas/oil conforms to the applicable requirements of either FIA or FMTL.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the currentversion before use.

3.4.1.5 OSHA. The supplier shall furnish a certificate of compliance, certifying that the tumblers being furnished under this specification conform to the applicable requirements for laundry equipment 29 CFR 1910.264 of OSHA Standard. The Government reserves the right to check test such tumblers to determine the validity of the certification.

3.5 Materials and components. Materials and components shall be as specified herein and on the applicable detail specification. Materials not definitely specified shall be of the quality normally used by the manufacturer for tumblers provided the completed items comply with all provisions of this specification.

3.5.1 Corrosion-resistant metal. Corrosion-resistant metal used in the fabrication of the tumblers shall be stainless steel conforming to requirements of ASTM Specification A167 or A176, as applicable; nickel-copper-alloy conforming to Composition A of QQ-N-281 or QQ-N-288; or copper alloy castings conforming to any copper alloy number 951-960 of QQ-C-390.

3.5.2 Materials for fasteners. Rivets, bolts, screws, nuts, and washers shall be of carbon or alloy steel, except where brass or corrosion-resistant metal is to be fastened, in which case they shall be of brass or corrosion-resistant metal, respectively (see 3.5.1). Where dissimilar metals are fastened, bolts, rivets, screws and nuts shall be of corrosion-resistant metal (see 3.5.1).

3.5.3 Pipe, tube, fittings, and valves. Pipe, tube, fittings, and valves used for water, steam, and to condensate lines shall comply with accepted plumbing practice.

3.5.4 Conduit, fittings, and electric wire. Electrical conduit shall be rigid, electrical metallic tubing, armored cable, or liquid-tight flexible conduit. All electrical wire shall be stranded copper, heat-resistant grade, thermoplastic insulated. Solid (nonstranded) wire shall not be used.

3.5.5 Motors. Motors shall be continuous, drip proof or splash proof type for operation in a 40 degree Centigrade (C) ambient temperature. All motors shall conform to the requirements of CC-M-636 or CC-M-641, as applicable, and to the following requirements:

(a) All motors shall have windings impregnated to resist moisture.

(b) Each motor shall have the proper starting characteristics, ample power, and adequate thermal dissipation capacity to comply with requirements specified herein, with a reasonable factor of safety, commensurate with the application and expected duty cycle to which each motor might be subjected under normal operating conditions.

(c) Motors shall be equipped with ball bearings, except that motors of 1/2 horsepower or less when used for horizontal applications may have sleeve bearings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

(d) Motor bearings shall be of permanently lubricated type or shall have adequate and accessible means for lubrication.

(e) Each motor of 1/8 horsepower rating or larger shall be provided with manually reset thermal-overload protection incorporated as a part of the motor or installed in the controller.

3.6 Design and construction. Design and construction of tumblers covered by this specification shall conform to the general requirements specified hereinafter and detail requirements in the applicable detail specification. Materials specified as corrosion-resistant metals shall conform to 3.5.1.

3.6.1 Design and general requirements. Tumblers shall be furnished complete with steam heating coils, or gas, oil, butane, propane, or gas/oil combination burner, as specified (see 6.2), motors, driving mechanism, starters, and controllers, including master switches, timers, temperature controls, and other electrical equipment and commercial accessories, as applicable. When tumblers are tested as specified in 4.8.3, the following general requirements shall be determined:

(a) The assembly of components shall have complete and proper interrelationship.

(b) Moving parts and bearings shall not overheat.

(c) Starting and stopping devices shall operate smoothly and positively.

(d) There shall be no structural failures.

3.6.1.1 Water evaporation. Water evaporation requirements as identified in the detail specifications shall be based on the evaporation rate when drying white terry cloth bath towels, conforming to requirements for type II, class 4, style A of DDD-T-551 having a 60 percent moisture content. The ambient temperature shall be 72.5 $\pm$ 2.5 F and the relative humidity shall be 50 $\pm$ 5 percent.

3.6.1.2 Frame. The frame shall be constructed of ferrous metal formed to provide a rigid support for the cylinder housing, heating unit, fan, and driving mechanism.

3.6.1.3 Housing. The cylinder, heating unit, and fan shall be inclosed in a ferrous metal housing. The housing shall be airtight and shall be provided with a tight fitting door or removable panels giving ready access to parts requiring maintenance.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the current version before use.

3.6.1.4 Door(s). The tumbler shall be provided with a loading door (also unloading door, when applicable). The door(s) shall be interlocked with the motor to automatically shut-off the motor when any tumbler door is opened. On gas, oil, butane, propane, heated tumblers, the door(s) shall be interlocked with burner controls to prevent ignition until door is in "closed" position.

3.6.1.5 Cylinder. The cylinder shall be constructed of galvanized ferrous metal or other metal having comparable corrosion resistance and strength. The cylinder shall be fabricated of metal sheets containing staggered perforations in accordance with the manufacturer's standard product. The cylinder shall contain lifting ribs fabricated of galvinized ferrous metal and the quantity and method of mounting shall be as specified in the detail specification.

3.6.1.6 Clothing protection. The gap between the open-end(s) of the cylinder and the door frame shall be constructed to prevent damage to clothing.

3.6.1.7 Heat conditioning system. Each machine shall be fitted with a suitable housed heat conditioning system containing a steam or gas, oil, propane, butane, or gas/oil heater (as specified in the detail specification), an electric motor-driven blower assembly, a manually operated tumbler temperature control, and a motor controller. Openings of ample size shall be provided for exhausting the air after passage through the tumbler. The openings shall be provided with duct connections.

3.6.1.8 Steam heating coils. Steam heating coils of the extended surface type shall be of copper or alloy construction for the specified steam working pressure of 3.6.1.22. The number of coils shall be sufficient to enable the tumbler to evaporate the quantity of water specified in the detail specification. The coils shall be divided into two or more sections and each section shall have a steam supply and drain. Inlet and return valves shall be provided so as to permit the by-passing of one or more sections, when desired, and connecting returns to a common outlet. The by-pass valve assembly may be shipped detached, with instructions for assembly in the field. Coil headers, if required, shall be fabricated from malleable iron, bronze or steel.

3.6.1.9 Burner heater. The burner heater for tumblers shall be direct heated gas, oil, butane, propane, or gas/oil, as specified (see 6.2). Each fired system shall be of sufficient capacity to evaporate the quantity of water specified.

3.6.1.10 Drive. Drive shall be provided by one or more motors of ample capacity to operate the cylinder and fan. Means shall be provided for adjustment if connections involve belts or chains.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the current version before use.

3.6.1.11 Lint filter. All tumblers shall be provided with a means for filtering lint for the exhausting air.

3.6.1.12 Applicability of electrical characteristics. The electrical parts of machines shall be designed for the voltage, frequency, and number of phrases specified in the invitation for bids (see 6.2). Operational and functional testing shall be performed at the voltage, frequency, and number of phases specified.

3.6.1.13 Controllers. Each motor shall be provided with magnetic starters and thermal motor overload protection. The tumbler shall have a timer, limit switches, remote control pushbutton switches, signals, indicating lights, and other controllers necessary to perform the intended functions. Unless otherwise specified, the controller shall be of the magnetic across-the-line type having a separate pole for each ungrounded conductor. Controllers for direct-current motors of more than 1/2 horsepower shall be of the reduced-voltage type having one or more steps of armature resistance for smooth acceleration. The timer shall be a mechanical device capable of timing a maximum cycle of 60 minutes for steam heated tumblers and 30 minutes for gas heated tumblers. The timer shall indicate completion of operation by an audible and visual signal. Pushbutton and timer control stations shall be mounted convenient to the operators. Pushbuttons shall be fully protected or recessed to prevent accidental operation. The drying fan and tumbler motor controls shall be wired through the door switch so that whenever the door is opened, operation of the drying air fan and tumbler will cease, when tested as specified in 4.8.4. The drying tumbler controls shall operate so that with the door closed and the timer on, the motor(s) must be started by pressing the motor "start" pushbutton and stopped by pressing the motor "stop" pushbutton. When the drying tumbler is not being used to dry clothes and the door is closed, all equipment operation will be stopped by pressing the motor "stop" button. The motor control circuit shall be wired to operate on a 120 volt system. Reversing tumblers shall revolve approximately an equal number of revolutions in either direction. Reversing tumblers shall be the same as non-reversing except that an electrical device shall be provided to reverse the cylinder at regular intervals.

3.6.1.14 Wiring and instructions. Wiring shall be installed in conduit and provided between all components mounted on the tumbler and shall be of the materials specified in 3.5.4, as applicable, except flexible metallic conduit may be used where required for motors with belt adjustment. All wiring of electrical components shall be completed so there is only one point of connection to the outside power source. Complete wiring and connection diagrams of the tumbler including instructions for connecting wall-mounted controllers shall be furnished with each tumbler.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

3.6.1.15 Safety guards. All exposed belts, chains, shafts, pulleys, gears, and other moving parts shall be guarded. Guards shall be cast iron, sheet metal, or wire mesh, securely fastened and readily removable. Guards shall allow access holes for parts requiring frequent adjustment and lubrication and these access holes shall be provided with covers.

3.6.1.16 Bases and foundation. Tumbler bases shall be constructed so that points of contact with the foundation are in the same horizontal plane. Provisions shall be made for anchoring the machine to the foundation to keep the contact surfaces of the bases in the same plane as the contact surface of the floor during full load operation. All required channels, I-beams, anchor bolts and other foundation materials, except concrete, and a diagram of recommended foundation details shall be furnished with each tumbler.

3.6.1.17 Gearing. All gears shall have machined teeth and shall operate quietly. Spiral, hypoid, bevel, and worm gears, if used, shall operate in a lubricant. When run in an oil bath, the pinion shall be of ferrous metal and the gear shall be of ferrous metal or bronze. Gears not run in an oil bath may be of phenolic formaldehyde resin, or other noise-reducing material. Where gears are of composition material, either the motor pinion or gear with which it meshes shall be at least 1/4 inch wider than the other. When thrust bearings are used on the armature shaft, the gear and pinion may be of the same width.

3.6.1.18 Lubrication. Provision shall be made for lubrication of all moving parts. Forced lubrication fittings shall be provided and be accessibly located for lubricating chain drives and all bearings, except on motors (see 3.5.5), where lubricant reservoirs are required, and where lubrication pressures may damage seals or other parts. Where lubricant reservoirs are required for chain or gear drives, the reservoirs shall be provided with a threaded cap at the top and drain plug at the bottom.

3.6.1.18.1 Seals. Means shall be provided to prevent the lubricant from entering parts of the machine where clothes are handled and shall prevent moisture from entering the bearings.

3.6.1.19 Chains, belts, and pulleys. Belts shall be of the grade commercially known as multiple-drive or variable speed types. When a single belt is furnished with a fractional-horsepower motor, a belt commercially known as a fractional-horsepower belt may be supplied. Pulleys shall be grooved for the belts used. Driving chains shall be of the silent or roller-chain type.

3.6.1.20 Belt and chain tension adjustment. Means shall be provided for readily adjusting belt and chain tension.

3.6.1.21 Piping steam. Necessary integral pipe, fittings, condensate traps, pressure reducing valves, and all other valves for steam and drains shall be installed or furnished as complete assemblies suitable for installation with unions or other standard fittings. All steam lines shall be installed to comply with accepted plumbing practice. Installation instructions shall be furnished with uninstalled assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

3.6.1.21.1 Valves and traps, steam. Steam, drain, condensate (with three-valve-by-pass and trap for each heating coil), and other operational valves shall be so installed that they are readily accessible for operation and maintenance.

3.6.1.22 Working pressure, steam. Assemblies of pipes, tubes, fittings, and coils using steam shall be designed for a 125 pounds per square inch gage (psig) steam working pressure. Heating coils shall be capable of withstanding 250 psig hydrostatic test without any leaks, when tested as specified in 4.8.1.

3.6.1.23 Piping. Pipe, fittings, valves and traps shall be installed or furnished as complete assemblies. Installation instructions shall be furnished with uninstalled assemblies.

3.6.1.24 Working pressures, fuel. Assembly of fuel piping, valving, and fittings, when tested as specified in 4.8.2 at the rated input of the equipment furnished, shall have a gas pressure (inches of water column) compatible with the BTU's as stated by the manufacturer for gas, oil, butane, or propane burner operation.

3.6.1.25 Name and data plates. Tumblers for Military procurement shall be marked for identification in accordance with MIL-STD-130 on a plate conforming to MIL-P-514. Tumblers for civil agency procurement shall be marked in accordance with the manufacturer's standard practice.

3.6.1.26 Finish. Unless otherwise specified (see 6.2), the tumblers, except stop pushbuttons, shall be finished according to the manufacturer's standard commercial practice. Emergency stop and normal stop pushbuttons shall be red and this color shall be limited to stop pushbuttons.

3.6.1.27 Electromagnetic compatibility. When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.8.6).

3.6.1.28 Safety devices. Tumblers shall be equipped with the following safety devices which shall be operative when tested as specified in 4.8.4.

(a) Lights to designate that tumbler is in operation (200 and 400 pound size only).

(b) Manual override device to permit opening door in case of power failure.

In addition to the above devices, fuel heated tumblers shall be equipped with the following safety devices which shall be operative when tested as specified in 4.8.4.

(a) Excess temperature cut-out switches and operating temperature controls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

(b) Excess temperature sprinkler protection (200 and 400 pound size only).

(c) Audible and visual alarms that indicate interruption to the safety control circuit.

(d) Untraviolet sensor of continously checking type which shall operate in conjunction with an electronic relay to close fuel valve 2-4 seconds after flame propagation in the combustion space ceases (200 and 400 pound size only).

(e) Flame sensing rod, if ultraviolet sensor is not furnished.

3.7 Health and safety compliance. The item(s) furnished under this specification shall comply with the applicable standards promulgated under the Occupational Safety and Health Act.

3.8 Workmanship. Tumblers shall be free from defects such as fractures, splits, punctures, tears, dents, creases, or malformation. There shall be no sharp edges, slivers, burrs, projections or other defects. The inside of the cylinder shall be smooth and free of burrs, sharp edges, and other imperfections.

3.8.1 Application of finish. The finish applied to the end item shall be continuous, smooth, adherent without discoloration or foreign material imbedded and contain no sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores, pits, lumps, flux or orange peel. No rust, rough grinds, or toolmarks shall show through the coating.

3.8.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, smooth and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux (when flux is used) shall be removed from the finished weld area.

3.8.3 Soldering. Soldering shall be complete and adherent with all flux and residue removed and shall contain no pinholes.

3.8.4 Bolts, nuts, screws, studs, and other threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.8.5 Rivets. Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the reveted members.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:00Z
Check the source to verify that this is the current version before use.

3.8.6 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wire nuts shall not be used and solderless connectors shall have insulation grip. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. The supplier shall furnish a certificate of compliance, certifying that the items being furnished under this specification conform to the applicable standards of OSHA. The Government reserves the right to check test such items to determine the validity of the certification.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required, the tumbler shall be visually examined for defects listed in table I (and the applicable detail specification) specified dimensions and tested as specified in 4.8.1, 4.8.2, 4.8.4, 4.8.5, and when applicable 4.8.6. The presence of a visual or dimensional defect, or failure of any test shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.5.2 End item inspection. The inspection lot shall be all tumblers offered for inspection at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

4.5.2.1 Visual inspection. The tumbler shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I - Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not continuous, smooth, adherent		X
	Sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores, pits, lumps, flux, or orange peel		X
	Rust, rough grinds or toolmarks, foreign matter imbedded, and not adherent		X
Materials	Materials not as specified	X	
Construction and workmanship, general (applicable to all components and assemblies)	Any component fractured, split, punctured, dented, or deteriorated	X	
	Any component not in proper alignment	X	
	Sharp burr, edge, sliver, or splinter	X	
	Steam heating coils not divided in two or more sections, as specified	X	
	Number and construction of cylinder ribs not as specified per detail sheets	X	
	Lint filters (screens or collectors), not as specified	X	
	Component loose		X
	Not connected or joined as specified	X	
	Any adjustable assembly that cannot be adjusted to perform the function intended	X	
	Any component not readily accessible for required servicing	X	
Welding and brazing (where required)	Not sound or smooth: not free of porosity of cracks: missing; incomplete	X	
	Slag inclusion, slight undercut, scale, or flux deposits not removed		X
Soldering (when applicable)	Missing, not adherent or incomplete	X	
	Not clean (flux or flux residue not removed), not smooth (surface not finished neatly), or pinholes in solder		X

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the current version before use.

TABLE I - Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, broken, stripped, or fractured	X	
	Loose		X
Rivets (when applicable)	Broken, loose, not peened, or insuffieiently peened	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed in conduit where required	X	
	Adequate slack not provided for wiring to relieve strain	X	
	Excessive insulation stripped from wiring	X	
	Solenoid operated valve, temperature control, signal light, timing device and audio signal inoperative, missing, or not as specified (when applicable)	X	
Insulation	Missing	X	
	Not permanently attached	X	
Lubrication (where applicable)	Not lubricated where required	X	
	Lubrication fitting not accessible for servicing	X	
Marking for identification	Missing, incomplete, not legible		X

4.5.2.2 Dimensional examination. The tumbler shall be examined for compliance with dimensions specified. Any dimension not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.5.2.3 End item testing. When a first article is not required, the first produced tumbler shall be examined and tested as specified in 4.8.4, 4.8.5, and 4.8.6, if applicable. Each sample tumbler shall be tested as specified in 4.8.1, 4.8.2, and 4.8.3, as applicable. Failure to pass a test shall be cause for rejection of the inspection lot. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed as defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

4.6 Code and standard examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.7 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with MIL-L-3153.

4.8 Test.

4.8.1 Hydrostatic pressure test. All assemblies of pipe, tubing, fittings, and coils using steam shall be tested with a 125 psig steam pressure. Heater coils shall be tested with 250 psig hydrostatic pressure. Test pressures shall be maintained for 15 minutes to determine conformance with 3.6.1.22. Any nonconformance shall constitute failure of this test.

4.8.2 Fuel pressure test. Fuel units shall be tested with a water column manometer for 10 minutes to determine conformance with 3.6.1.24 for pressures as stated by the manufacturer. Any nonconformance shall constitute failure of this test.

4.8.3 Operation testing. Each tumbler furnished under this specification shall be connected to power and operationally tested without garments or laundry. In the course of testing for operational requirements specified in 3.6.1 and in the detail specifications, determination shall be made to establish compliance with all performance requirements applicable to the fully assembled item except those that require functional testing such as processing of garments or laundry through the equipment. Any nonconformance shall constitute failure of this test.

4.8.4 Functional testing. The tumbler shall be tested by making the necessary connections, such as electrical power and steam or appropriate fuel, as applicable, and by operating under full service load as established in the applicable paragraph of the detail specification. Any nonconformance shall constitute failure of the functional test for the tumbler specified.

4.8.4.1 Test items, steam and fuel. The following items, as applicable to the tumbler being tested, shall be checked for compliance with the requirements specified in the general and applicable detail specification:

(a) Size of machine (capacity)
(b) Construction of frame
(c) Construction and positioning of cylinder
(d) Direction(s) of rotation of cylinder
(e) The method of driving and any adjustment of the drive mechanism
(f) Door interlock so that tumbler will operate only with door closed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

(g) Motor Control
(h) Timing device
(i) Amount of water evaporation
(j) Light to indicate that tumbler is operating
(k) Manual override device to permit opening the loading door in case of power failure
(l) Construction of steam heating coils or fuel heater
(m) For fuel heated design, excess temperature cut-out switches and operating temperature controls
(n) For fuel heated design, excess temperature sprinkler protection
(o) For fuel heated design, audible and visual alarms that indicate interruption of the safety control circuit
(p) For fuel heated design, ultraviolet sensitive flame scanner or flame rod with controls to provide flame failure control of the pilot and main burner

4.8.5 Surface temperature. At an ambient temperature of 80 F., the tumbler shall be operated at the maximum temperature setting for three complete drying cycles. Using a pyrometer, twenty readings shall then be taken on each side, back, and top surface (total of 80 readings). The pyrometer settings shall be spaced approximately 18 inches apart and 18 inches from any edge. When more than 15 readings from all surfaces are in excess of 115 F., the total area of excessive heat (116 to 200 F.) shall be calculated and the area percentage obtained by dividing total machine surface area into the excessive heat surface area. When the area of excessive heat is greater than 15 percent of the total area or when any reading is greater than 200 F., the machine shall be rejected.

4.8.6 Electromagnetic compatibility. When electromagnetic compatibility is required, the first article or initial unit of production, as applicable shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report, and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each complete tumbler shall be preserved, packaged, packed, and marked in accordance with the applicable requirements of MIL-L-3153. Preservation and packaging shall be level A or C and packing shall be level A, B, or C (see 6.2). Drying tumblers shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, or disassembly results in lower transportation cost to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

5.2 Marking for civilian agencies. Shipping containers shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The laundry tumblers covered by this specification are intended for use in fixed installations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification and applicable detail specification.

(b) Types, styles, reversibility, sizes, and kind of heat required (see detail specification).

(c) First article. When a first article is required (see 3.2, 4.4 and 6.3).

(d) The type of fuel or combination thereof that is required (see 3.6.1).

(e) Electrical characteristics required (voltage, phase, and frequency) (see 3.6.1.12).

(f) Finish when other than manufacturers' standard is required (see 3.6.1.26).

(g) When electromagnetic compatibiltiy is required (see 3.6.1.27).

(h) Levels of preservation and packaging, and level of packing required (see 5.1).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

6.4 Successful commercial operation. No item of equipment offered in response to invitation for bids will be acceptable unless the manufacturer has had equipment of the specific type and performing the required operations and end functions as set forth herein operating successfully in a commercial or institutional laundry for at least twelve full months prior to the date of the invitation for bids. Equipment installed in any location for test purposes will be considered as not in compliance with this requirement.

6.5 Supersession data. Changes in classification of the tumblers between this revision of the specification and the previous edition are as follows:

Old Designation	New Designation
Type I, Style B	00-D-1297/1
Type I, Style B, Size 30	Deleted
Type I, Style C	00-D-1297/2
Type I, Style D	Deleted
Type II, Style A	Deleted
Type II, Style B	00-D-1297/3
Type II, Style C	00-D-1297/4
Type II, Style D	00-D-1297/5

Custodians:

Army - GL
Navy - YD
Air Force - 84

User activities:

Army - ME, CE
Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

Project No. 3510-0201

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 20 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: Drying and Conditioning Tumblers, Laundry, Commercial: General Specification For 00-D-1297B/GEN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.